

Extrapolating Simualated Gravitational Wave Data to Infinity for the LIGO project

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 - Extraction
 - Perturbative Techniques
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 - Cauchy Characteristic Evolution

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 - $\Psi_4 = C_{\alpha\beta\gamma\delta} l^\alpha \bar{m}^\beta n^\gamma \bar{m}^\delta$
 - Applications and uses in simulation

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 - Applications and uses in simulation
- Obtaining h from simulated Ψ_4
 - The Wave Zone approximation: $\Psi_4 = \frac{\partial^2}{\partial t^2}(h_+ - ih_\times)$
 - Conditions:
 - Far away from the source of GW (asymptotically flat spacetime)

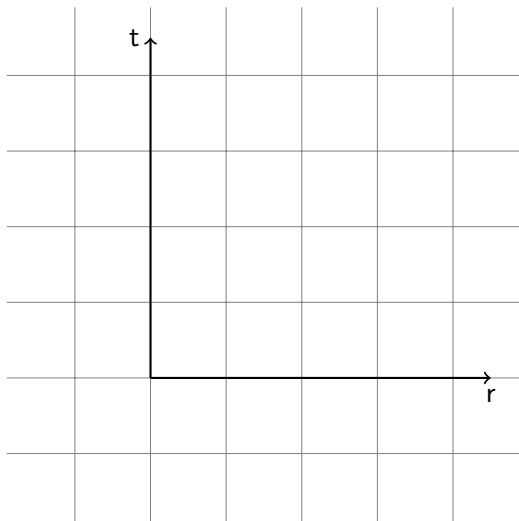
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- The LIGO Detector
 - How it works: measuring strain
 - Deciphering noise from meaningful data

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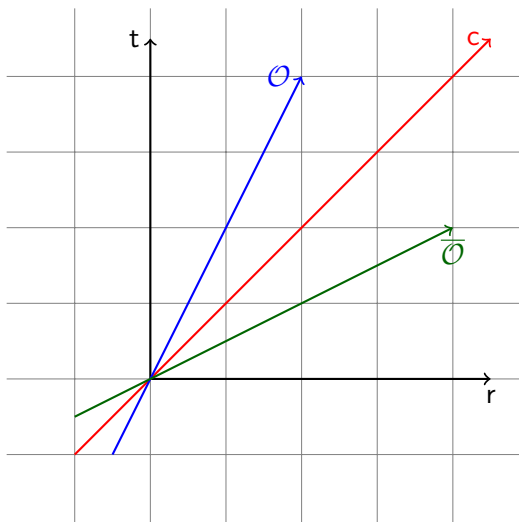
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Spacetime Diagrams



- The coordinates have different increments: t is measured in light-meter (amount of time it takes light to travel 1 meter)
- Spacetime diagrams of flat space have infinities that cannot be pictured
- Need a different coordinate system

Spacetime Diagrams (cont.)



- Green denotes space-like curve
- Blue denotes time-like curve
- Red is the curve followed by light, also called null-like curve (observe this has an angle of 45°)

Penrose Diagram

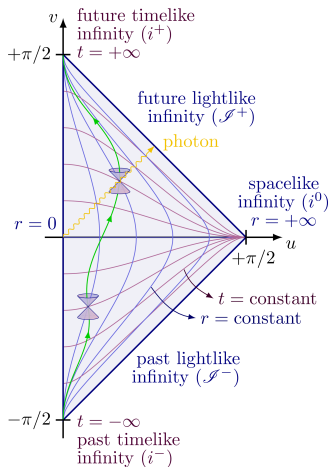


Figure: Penrose Diagram

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- Numerical relativity (NR) are simulations of certain systems governed by the laws of GR (the Einstein Field Equations)
- These equations have no analytical solution in the case of BH-BH collisions, so they have to be computed numerically on super computers

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

But how does NR work?

Numerical Relativity

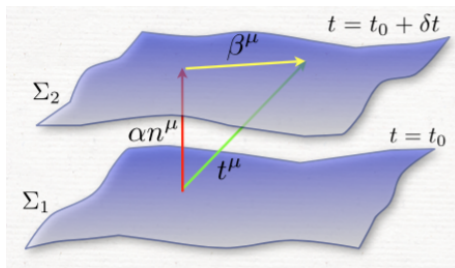
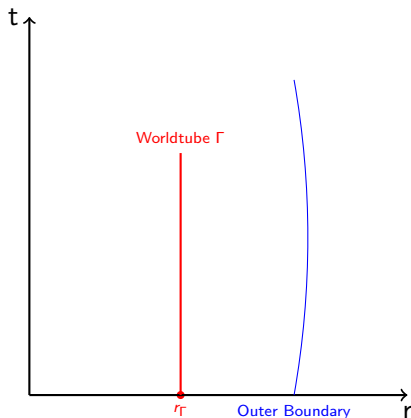


Figure: Sample time evolution

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Extracting Ψ_4



- Set up detectors somewhere within the simulation (called worldtube)
- Use the detectors to obtain Ψ_4
- Use the approximation $\Psi_4 \approx \ddot{h}$ to obtain strain
- Easiest but also most error-prone

Perturbative Techniques to obtain h

- Similar to the extraction technique
- Key difference: use formula developed from perturbation theory to approximate Ψ_4 at $\mathcal{I}^+$.

$$r \psi_4^{\ell m} \big|_{r=\infty} = r \psi_4^{\ell m}(t, r) - \frac{(\ell-1)(\ell+2)}{2r} \int dt [r \psi_4^{\ell m}(t, r)] \\ + \frac{(\ell-1)(\ell+2)(\ell^2 + \ell - 4)}{8r^2} \int \int dt dt [r \psi_4^{\ell m}(t, r)] - \frac{3M}{2r^2} \int dt [r \psi_4^{\ell m}(t, r)] + \mathcal{O}(1/r^3).$$

Perturbative formula for Ψ_4 at $\mathcal{I}^+$

- This formula holds when the resulting BH is non-spinning (inaccurate when they do spin)

Extrapolating Ψ_4 to $\mathcal{J}^+$

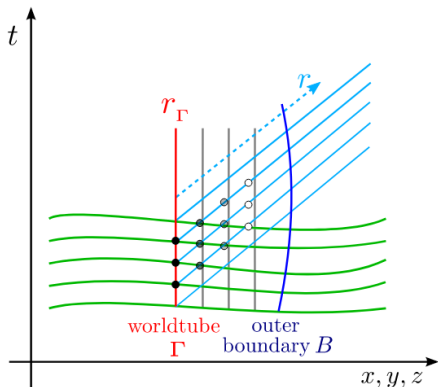


Figure: Extrapolative Technique to obtain h

Bishop, N.T., Rezzolla, L. (2016)

- First, since Ψ_4 is a waveform, we can break it down as $\Psi_4^{l,m}(T, R) = A^{l,m} e^{i\phi^{l,m}}$
- Then, to A and ϕ we can apply the polynomial approximation

$$A(T, R) \approx \sum_{k=0}^n A_k(T) \left(\frac{1}{R} \right)^k$$

(similarly for ϕ)

- Setup multiple detectors at different radii (r) and obtain A , ϕ at those radii

Cauchy Characteristic Evolution

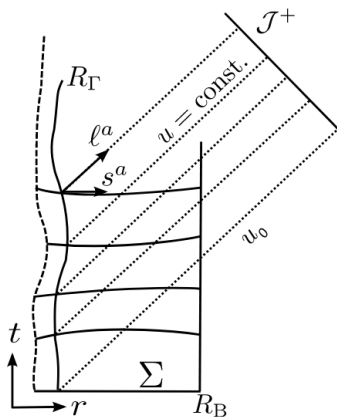


Figure: Cauchy Characteristic Evolution

Taylor, et al. (2018)

- Obtain metric data on the worldtube
 - Grab the metric and it's derivatives on world tube $\{g_{xx}, g_{xy}, \dots, \partial_t g_{xx}, \partial_t g_{xy}, \dots\}$
 - Constitutes the "Cauchy" in Cauchy Characteristic Evolution
- Evolve this metric data on Characteristic hypersurfaces
 - Defined as the null-like dashed lines
 - Denotes Characteristic evolution
- Different groups studying gravitational waves will follow different schema

- Bishop, N.T., Rezzolla, L. Extraction of gravitational waves in numerical relativity. Living Rev Relativ 19, 2 (2016).
<https://doi.org/10.1007/s41114-016-0001-9>
- H. Nakano, et al. “Perturbative extraction of gravitational waveforms generated with Numerical Relativity”. Physical Review D, Published + Submitted, vol. 91, no. 10, American Physical Society, Dec. 2015, p. Art. No. 104022,
<https://link.aps.org/doi/10.1103/PhysRevD.91.104022>
- Taylor, N. W., et al. “Comparing Gravitational Waveform Extrapolation to Cauchy-characteristic Extraction in Binary Black Hole Simulations”. Physical Review D, Published + Submitted, vol. 88, no. 12, American Physical Society, Dec. 2013, p. Art. No. 124010, doi:10.1103/PhysRevD.88.124010.